

Kantorovich formulation of quadratic optimal transport

$$W_2^2(p, q) = \inf_{\pi \in \Pi(p, q)} \int_{\mathbb{R} \times \mathbb{R}} |x - y|^2 d\pi(x, y)$$

$$\Pi(p, q) = \{\pi : (pr_1)_\# \pi = p(x) dx, (pr_2)_\# \pi = q(y) dy\}$$

